

Scalable Optimal Margin Distribution Machine

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Abstract

Optimal margin Distribution Machine (ODM) is a newly proposed statistical learning framework rooting in the latest margin theory, which demonstrates better generalization performance than the traditional large margin based counterparts. However, it suffers from the ubiquitous scalability problem regarding both computation time and memory storage as other kernel methods. This paper proposes a scalable ODM, which can achieve nearly ten times speedup compared to the original ODM training method. For nonlinear kernels, we put forward a novel distribution-aware partition method to make the local ODM trained on each partition be close and converge fast to the global one. When linear kernel is applied, we extend a communication efficient SVRG method to accelerate the training further. Extensive empirical studies validate that our proposed method is highly computational efficient and almost never worsen the generalization.

1 Introduction

Recently, the study on margin theory [Gao and Zhou, 2013] demonstrates an upper bound disclosing that maximizing the minimum margin does not necessarily result in a good performance. Instead, the distribution rather than a single margin is much more critical. Later on, the study on lower bound [Grønlund *et al.*, 2019] further proves that the upper bound is almost optimal up to a logarithmic factor. Inspired by these insightful works, Zhang and Zhou [2019] propose the *Optimal margin Distribution Machine* (ODM), which explicitly optimizes the margin distribution by maximizing the mean and minimizing the variance simultaneously and exhibits much better generalization than the traditional large margin based counterparts. Due to the superiority shown on both binary and multi-class classification tasks, many works attempt to extend ODM to more general learning settings, just to list a few, cost-sensitive learning [Zhou and Zhou, 2016; Cheng *et al.*, 2017], weakly supervised learning [Zhang and Zhou, 2018a; Zhang and Zhou, 2018b; Luan *et al.*, 2020;

Zhang and Jin, 2020; Cao *et al.*, 2022], multi-label learning [Tan *et al.*, 2020; Cao *et al.*, 2021], online learning [Zhang *et al.*, 2020], and regression [Rastogi *et al.*, 2020]. Plenty of successes on various learning tasks validate the superiority of this new statistical learning framework. However, with the dramatic progress of digital technologies, the data generated devices become as diverse as computers, mobile phones, smartwatches, cars, etc., and the amount of data created each day grows tremendously, thus these ODM based extensions suffer from the scalability problem regarding both computation time and memory storage as other kernel methods.

There have been many works devoted to accelerating kernel methods, which can be roughly classified into three categories. The first category is based on approximation, e.g., the random Fourier feature [Rahimi and Recht, 2007] takes the trigonometric functions as basis functions to approximate the kernel mapping, the Nyström method [Williams and Seeger, 2001] generates a low-rank approximations by sampling a subset of columns, and the coresset [Tan *et al.*, 2019] adaptively sketches the whole data by choosing some landmark points. The second category divides the data into partitions on which local models are trained and combined to produce a larger local or global model, e.g., in [Graf *et al.*, 2004; Hsieh *et al.*, 2014; Singh *et al.*, 2017], a tree architecture on partitions is designed first, guided by which the solutions of different partitions are aggregated; in [Yu *et al.*, 2005; Navia-Vazquez *et al.*, 2006; Loosli *et al.*, 2007], the key instances identification and exchange are further introduced to accelerate the training; in [Si *et al.*, 2017], both low-rank and clustering structure of the kernel matrix are taken into account to get an approximation of kernel matrix. The third category is directly applying the distributed-style optimization method, such as the augmented Lagrangian method [Forero *et al.*, 2010] and the alternating direction method of multipliers [Boyd *et al.*, 2010], or extending existing solver to a distributed environment, e.g., distributed SMO [Cao *et al.*, 2006].

Notice that the random Fourier feature adopts a data-independent kernel mapping and the Nyström method takes a data distribution-unaware sampling, hence their performance are both inferior to the coresset method [Tan *et al.*, 2019], which inspires us to leverage data as heavily as possible. Moreover, the distributed off-the-shelf *quadratic programming* (QP) solvers can be directly applied to train ODM, but

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